

1. (a) False

The 2-D field $\vec{F} = \langle P, Q \rangle$, where $P = \frac{y}{x^2+y^2}$, $Q = \frac{-x}{x^2+y^2}$ has

$$\text{curl } \vec{F} = (Q_x - P_y) \hat{k} = \vec{0}.$$

Yet, if $C: \vec{r}(t) = \langle \cos t, \sin t \rangle$, $0 \leq t \leq 2\pi$ (unit circle centered at $(0,0)$ oriented counterclockwise,

$$\oint_C \vec{F} \cdot d\vec{r} = -2\pi,$$

not zero. (See Example A linked from Ch.6.5 entry in MOM.)

The statement can be modified as in Theorem 6.18, which implies that

$\oint_C \vec{F} \cdot d\vec{r} = 0$ when C is a closed curve in a simply connected domain where $\text{curl } \vec{F} = \vec{0}$.

(b) False

$\text{div}(\text{curl } \vec{F})$ is a scalar function.

(c) False

(d) True

(e) False

Generally, when $\vec{F} = \langle P, Q, R \rangle$, then

$$\text{curl } \vec{F} = \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle,$$

so

$$\begin{aligned} \text{curl}(\text{curl } \vec{F}) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ R_y - Q_z & P_z - R_x & Q_x - P_y \end{vmatrix} = \begin{bmatrix} \frac{\partial}{\partial y}(Q_x - P_y) - \frac{\partial}{\partial z}(P_z - R_x) \\ -\left[\frac{\partial}{\partial x}(Q_x - P_y) - \frac{\partial}{\partial z}(R_y - Q_z)\right] \\ +\left[\frac{\partial}{\partial x}(P_z - R_x) - \frac{\partial}{\partial y}(R_y - Q_z)\right] \end{bmatrix} \\ &= \langle Q_{xy} - P_{yy} - P_{zz} + R_{xz}, P_{yx} - Q_{xx} - R_{yz} + Q_{zz}, P_{zx} - R_{xx} - R_{yy} + Q_{zy} \rangle, \end{aligned}$$

whose components are generally nonzero.

(f) True (See Theorem 6.16, Section 6.5)

2. (a) Any open connected region Ω of the plane that includes $(0,0)$, where $\vec{F}$ is undefined, naturally results in $\vec{F}$ not having continuous derivatives throughout

Ω . If the open connected region Ω does not include $(0,0)$, then $\vec{F}$ has continuous derivatives $(\partial P/\partial x, \partial P/\partial y, \partial Q/\partial x, \partial Q/\partial y)$ within Ω .

(b) Call the region Ω' after a single point is removed from Ω . If Ω is an open, connected, and simply connected region of the plane then

Ω' is still open

Ω' is still connected

Ω' is no longer simply connected

3. (a) The components $P = 6xy$, $Q = 3x^2 - ye^y$ are continuous on the entire plane (an open, connected and simply connected region), and

$$Q_x = 6x = P_y, \quad \text{so} \quad \text{curl } \vec{F} = (Q_x - P_y) \hat{k} = \vec{0}.$$

So, $\vec{F}$ is conservative. Its potential function $f(x,y)$ satisfies

$$1^{\text{st}} \text{ requirement: } \frac{\partial f}{\partial x} = 6xy \Rightarrow f(x,y) = \int 6xy \, dx = 3x^2 y + g(y).$$

$$2^{\text{nd}} \text{ requirement: } \frac{\partial f}{\partial y} = 3x^2 - ye^y.$$

$$\text{But } \frac{\partial}{\partial y} [3x^2 y + g(y)] = 3x^2 + g'(y) \Rightarrow g'(y) = -ye^y$$

$$\Rightarrow g(y) = -\int ye^y \, dy = e^y(1-y) + C$$

Putting this together,

$$f(x,y) = 3x^2 y + e^y(1-y) + C$$

is a potential for $\vec{F}$ for every choice of constant C .

(b) Again, $\vec{F}$ has components defined for all of space, and at each (x,y,z) , $\text{curl } \vec{F} = \vec{0}$. So, $\vec{F}$ is conservative. For any potential f ,

$$\frac{\partial f}{\partial x} = 2xy + z^2 \Rightarrow f(x,y,z) = \int (2xy + z^2) \, dx = x^2 y + xz^2 + g(y,z).$$

This form for f has y -partial derivative

$$\frac{\partial f}{\partial y} = x^2 + 0 + \frac{\partial g}{\partial y}(y,z), \quad \text{required to be } x^2 + 2yz.$$

Comparing results, we see

$$g_y(y,z) = 2yz \Rightarrow g(y,z) = \int 2yz \, dy = y^2 z + h(z)$$

We can now revise our form of $f(x, y, z)$ to

$$f(x, y, z) = x^2y + xz^2 + y^2z + h(z).$$

Turning to the z -partial derivative,

$$f_z = 0 + 2xz + y^2 + h'(z), \text{ required to be } 2xz + y^2.$$

So, $h'(z) = 0$, implying the function $h(z)$ is a (true) constant C .

We have found all potential functions f have the form

$$f(x, y, z) = x^2y + xz^2 + y^2z + C.$$

4. Comparing $\vec{F}$ with 3(b), we have that it is conservative with potential

$$f(x, y, z) = x^2y + xz^2 + y^2z.$$

By the Fundamental Theorem of Line Integrals,

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= f(\text{final point}) - f(\text{initial point}) \\ &= f(3, -2, 3) - f(-1, 2, 1) = 21 - 5 = 16. \end{aligned}$$

$$5. \operatorname{div} \vec{F} = \frac{\partial}{\partial x}(3xyz) + \frac{\partial}{\partial y}(xye^z) + \frac{\partial}{\partial z}(-3xy) = 3yz + xe^z$$

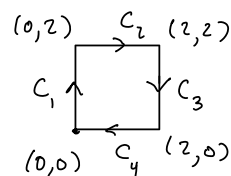
$$\operatorname{curl} \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3xyz & xye^z & -3xy \end{vmatrix} = \langle -3x - xye^z, 3xy + 3y, ye^z - 3xz \rangle.$$

6. (a) This is a circulation integral with $F = \langle 3xy, 2xy^2 \rangle$:

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= - \oint_C F \cdot d\vec{r} = - \int_0^2 \int_0^2 \left[\frac{\partial}{\partial x}(2xy^2) - \frac{\partial}{\partial y}(3xy) \right] dx dy \quad \left(\begin{array}{l} \text{Circulation -} \\ \text{Curl form of} \\ \text{Green's Thm.} \end{array} \right) \\ &= - \int_0^2 \int_0^2 (2y^2 - 3x) dx dy = - \int_0^2 \left(2xy^2 - \frac{3}{2}x^2 \Big|_0^2 \right) dy \\ &= - \int_0^2 (4y^2 - 6) dy = 6y - \frac{4}{3}y^3 \Big|_0^2 = 12 - \frac{32}{3} = \frac{4}{3}. \end{aligned}$$

Without employing Green's Theorem, we have

$$\oint_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} + \int_{C_3} \vec{F} \cdot d\vec{r} + \int_{C_4} \vec{F} \cdot d\vec{r}$$



$$\begin{aligned}
&= \int_0^1 \langle 3(0)(2t), 2(0)(2t)^2 \rangle \cdot \langle 0, 2 \rangle dt + \int_0^1 \langle 3(2t)(2), 2(2t)(2)^2 \rangle \cdot \langle 2, 0 \rangle dt \\
&\quad + \int_0^1 \langle 3(2)(2-2t), 2(2)(2-2t)^2 \rangle \cdot \langle 0, -2 \rangle dt + \int_0^1 \langle 3(2-2t)(0), 2(2-2t)(0)^2 \rangle \cdot \langle -2, 0 \rangle dt \\
&= \int_0^1 24t dt + \int_0^1 -32(1-2t+t^2) dt = 12t^2 \Big|_0^1 - 32 \left[t - t^2 + \frac{1}{3}t^3 \right]_0^1 \\
&= 12 - \frac{32}{3} = \frac{4}{3}, \quad \text{same as above.}
\end{aligned}$$

(b) C , here, is parametrized by $\vec{r}(x) = \langle x, \frac{1}{2}x \rangle$, $0 \leq x \leq 4$.

$$\begin{aligned}
\int_C x^2 dy + (2-3y)x dx &= \int_C \left[x^2 \frac{dy}{dx} + (2-3y)x \right] dx \quad \text{— following this maneuver, I insert } \frac{1}{2}x \text{ for } y, \frac{1}{2} \text{ for } \frac{dy}{dx}. \\
&= \int_0^4 \left(\frac{1}{2}x^2 + 2x - \frac{3}{2}x^2 \right) dx = x^2 - \frac{1}{3}x^3 \Big|_0^4 = 16 - \frac{64}{3} = -\frac{16}{3}.
\end{aligned}$$

(c) S is parametrized by $\vec{r}(x, y) = \langle x, y, x^2 - y \rangle$, and

$$\vec{r}_x \times \vec{r}_y = \langle -2x, 1, 1 \rangle \Rightarrow \|\vec{r}_x \times \vec{r}_y\| = \sqrt{1+1+4x^2} = \sqrt{2+4x^2}$$

$$\begin{aligned}
\text{So, } \iint_S xy^2 dS &= \int_0^4 \int_0^1 xy^2 \sqrt{2+4x^2} dx dy & u = 2+4x^2, \text{ so } du = 8x dx \\
&= \int_0^4 y^2 \int_2^6 \frac{1}{8} u^{1/2} du dy & 0 \leq x \leq 1 \Rightarrow 2 \leq u \leq 6 \\
&= \int_0^4 y^2 \left[\frac{1}{12} u^{3/2} \right]_2^6 dy = \frac{1}{18} (\sqrt{54} - \sqrt{2}) y^3 \Big|_0^4 \\
&= \frac{32}{9} (\sqrt{54} - \sqrt{2}).
\end{aligned}$$

(d) By the Divergence Theorem,

$$\begin{aligned}
\iiint_S \vec{F} \cdot \vec{n} dS &= \iiint_E \operatorname{div} \vec{F} dV = \iiint_E \left[\frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(2z^3) \right] dV \\
&= \iiint_E 6z^2 dV = \int_0^{2\pi} \int_0^{\sqrt{2}} \int_{r^2}^2 6z^2 r dz dr d\theta = 2 \int_0^{2\pi} \int_0^{\sqrt{2}} \left[z^3 \right]_{r^2}^2 r dr d\theta \\
&= 2 \int_0^{2\pi} \int_0^{\sqrt{2}} (8r - r^7) dr d\theta = 2 \int_0^{2\pi} \left[4r^2 - \frac{1}{8}r^8 \right]_0^{\sqrt{2}} d\theta = 12 \int_0^{2\pi} d\theta \\
&= 24\pi
\end{aligned}$$